Weather Prague

Multi-Source Weather Data Collection & Analysis

Authors: [Tibor Nemeth] [Tzu-Yun Lee]

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Project Overview

Build an automated pipeline that collects Prague weather and environmental data from three complementary sources, normalizes it into a common storage layer, and uses the combined dataset for historical trend analysis and weather model evaluation.

Each source serves a distinct purpose:

- **ČHMÚ** (Czech Hydrometeorological Institute) — real physical station measurements. The ground truth.
- **Open-Meteo** — modeled reanalysis data with 80+ years of history and variables that stations don't measure (solar radiation, soil temperature, UV index).
- **OpenWeatherMap** — a second model for cross-validation, plus air quality data (PM2.5, NO_2, O_3, CO, AQI) that the other sources don't provide.

Joined together on timestamp, the three sources produce a comprehensive environmental profile of Prague that no single source can provide alone.

Project Phases

	Phase	Focus	Effort
■	**Phase 1: Parse + Store**	Data collection, cleaning, normalization, storage	∼60%
■	**Phase 2: Analyze**	Historical trends, model accuracy, combined overview	∼30%
■	**Phase 3: Present**	Dashboard, notebook narrative, optional prediction demo	∼10%

The bulk of the project is Phase 1. Once the data is cleanly parsed and stored, analysis and presentation follow naturally from it.

Phase 1: Parse + Store (60% of effort)

Task 1: ČHMÚ — Station Measurements [hardest task]

Source: https://opendata.chmi.cz/meteorology/climate/
Format: Bulk CSV files, Czech column headers, station-coded filenames
Data: Temperature, humidity, wind speed/direction, precipitation, atmospheric pressure
Resolutions: 10-minute, hourly, daily (all three will be downloaded)
Scope: 10+ years of historical data for Prague station(s)

Purpose: ČHMÚ provides the only *real measurements* in the project — actual thermometers, rain gauges, and anemometers operated by the Czech national weather service. This is the

ground truth against which the two model-based APIs are compared. The 10-minute resolution data enables analysis of rapid weather changes (e.g. storm fronts, temperature drops) that hourly data would smooth out. Daily data provides the foundation for long-term climate trend analysis.

Parsing challenges:

- Directory structure must be crawled to find Prague-specific files
- Column headers are in Czech and must be mapped to English variable names
- Character encoding issues (Czech diacritics: ř, ž, č)
- Missing values may appear as `-999`, empty strings, or other sentinel values
- Timestamps may be in CET/CEST (local Czech time), requiring UTC conversion
- Station metadata must be cross-referenced to identify which station(s) are in Prague

Task 2: Open-Meteo — Historical Reanalysis + Live Forecast

Source:	https://api.open-meteo.com/v1/forecast (live) https://archive-api.open-meteo.com/v1/archive (historical)
Format:	JSON REST API, no authentication required
Data:	All basic weather variables + solar radiation, cloud cover, soil temperature, UV index
Resolution:	Hourly
Scope:	Historical backfill as far back as ČHMÚ data goes (for comparison), plus live collection going forward

Purpose: Open-Meteo serves two roles. First, as *enrichment*: it provides variables that ČHMÚ stations don't measure — solar radiation, UV index, soil temperature, detailed cloud cover. These complement the station data in the combined environmental overview. Second, as a *model to evaluate*: Open-Meteo's historical data is a mathematical reconstruction (ERA5 reanalysis), not a measurement. Comparing it to ČHMÚ's actual readings reveals how accurately the model represents Prague's weather.

Open-Meteo also returns *forecasts* (7–16 days ahead). The live collector stores these predictions so they can later be compared against what actually happened.

Parsing challenges:

- Historical backfill requires chunked requests (cannot request 10 years in a single call)
- Rate limiting on free tier — requests must be paced
- Null values in response arrays for certain hours/variables
- Precipitation is a sum over the preceding hour, not an instantaneous value — must not be confused with other instantaneous variables

Task 3: OpenWeatherMap — Cross-Validation + Air Quality

Source:	https://api.openweathermap.org/data/2.5/weather (current) https://api.openweathermap.org/data/2.5/air_pollution (AQI)
Format:	JSON REST API, API key required (free tier)
Data:	Temperature, humidity, wind, pressure, precipitation (weather) PM2.5, PM10, NO_2, O_3, CO, SO_2, AQI index (pollution)
Resolution:	Current snapshot (collected hourly by scheduler)
Scope:	Live collection only (free tier has no historical data)

Purpose: OWM serves two distinct purposes. First, as a *third weather source* for live cross-validation: when ČHMÚ says 22°C and Open-Meteo says 21°C, OWM provides a third opinion. Over weeks of collection, this builds a dataset for evaluating which model API is more accurate for Prague. Second, OWM's *air pollution endpoint* provides data that neither ČHMÚ nor Open-Meteo offer through their basic weather interfaces — PM2.5, ground-level ozone, nitrogen dioxide, and an overall Air Quality Index. This adds an environmental dimension to the combined Prague profile.

OWM also returns a 5-day forecast. Like Open-Meteo's forecast, these predictions are stored so they can later be compared to actual observations.

Parsing challenges:

- Response is nested JSON (not flat arrays) — must be flattened manually
- Weather and air pollution are separate endpoints with different response structures
- API key must be stored securely (environment variable / `.env` file, never committed)
- Free tier rate limits (60 calls/minute) — must handle 429 responses
- Snapshot timestamps must be rounded to the nearest hour for alignment with other sources

Task 4: Storage Layer

All three sources are stored in a single **SQLite database** (`data/weather.db`) with separate tables per source and resolution:

Table	Source	Resolution
`chmu_10min`	ČHMÚ	10-minute
`chmu_hourly`	ČHMÚ	Hourly
`chmu_daily`	ČHMÚ	Daily
`open_meteo_hourly`	Open-Meteo	Hourly
`owm_weather`	OWM	Hourly (from snapshots)
`owm_pollution`	OWM	Hourly (from snapshots)

Sources are kept in separate tables to preserve raw data integrity. Cross-source analysis is performed by joining tables on `timestamp_utc` at query time, not by merging at the storage layer.

Task 5: Automated Collection (GitHub Actions)

A scheduled GitHub Actions workflow runs every hour:

1. Fetch current weather from Open-Meteo and OWM
2. Fetch current air quality from OWM
3. Append new rows to the SQLite database
4. Commit the updated database back to the repository
5. Log success/failure with the Python `logging` module

ČHMÚ bulk data is downloaded separately (manually or on a weekly schedule) since it is published as periodic file dumps, not a real-time API.

Phase 2: Analyze (30% of effort)

Analytical objectives will be refined once the data is collected and explored. Initial directions include:

- **Historical climate trends** — temperature, precipitation, and extreme event analysis using ČHMÚ data (10+ years)
- **Model accuracy evaluation** — how well do Open-Meteo and OWM reproduce ČHMÚ's actual measurements? Which is more accurate for Prague?
- **Combined environmental overview** — joining temperature (from ČHMÚ), solar radiation (from Open-Meteo), and air quality (from OWM) into a single Prague environmental profile
- **Forecast accuracy tracking** — comparing stored predictions from Open-Meteo and OWM against subsequent actual observations

(This section will be expanded as the project progresses.)

Phase 3: Present (10% of effort)

- **Streamlit dashboard** — interactive exploration with date range filters, source comparison, and trend visualization
- **Jupyter notebook** — narrative-driven analysis highlighting key findings
- **Optional: prediction demo** — a simple model (e.g. linear regression or random forest) predicting next-day temperature, included only as a demonstration of the pipeline's ML-readiness

(This section will be expanded as the project progresses.)

Scope Boundaries

The following are explicitly **out of scope** to prevent scope creep:

- Production-grade ML forecasting (LSTM, Transformer, etc.)
- Multiple cities — Prague only
- Radar or satellite imagery processing
- Real-time alerting or notifications
- Deep air quality analysis (basic AQI inclusion only)

— end of project goals —